PSEUDO-COHERENCE ON PROJECTIVE SPACE IS DETECTED BY FINITELY MANY TWISTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Every perfect complex is pseudo-coherent and the tensor product of two pseudo-coherent complexes is pseudo-coherent. used at proof lines 8797-8806. Gives the implication from the first condition to the second and the trivial implication from the second to the third.
- The direct image of a pseudo-coherent complex along a flat proper morphism of finite presentation is pseudo-coherent. invoked at proof lines 8804-8806. Produces the second condition from the first.
- The derived category of quasi-coherent modules on projective space is equivalent to the derived category of modules over the graded algebra of the projective space, with the generator corresponding to the structure sheaf. invoked at proof lines 8810-8820. Lets the problem be translated into a statement about modules over a ring.
- The derived Hom over the ambient algebra computes the graded pieces of the tensor product with the generator, and pseudo-coherence descends along the equivalence. invoked at proof lines 8821-8862. Concludes that the given complex is pseudo-coherent from the condition on the twists.

Source reference. Original section 0B6A: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 8794-8796 to identify the two expressions and thus the equivalence of conditions (3) and (4); the deduction is inside the counted span.

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 8798-8799 to finish the equivalence of conditions (3) and (4); everything else used for it is inside the counted span.

Source reference. Original lemma 09J3: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 8801-8803 to see that the tensor product with a pseudo-coherent object is pseudo-coherent; the implication from the first to the second condition is inside the counted span.

Source reference. Original lemma 0CSD: sources/stacks-perfect/pinned-original-source.tex, source lines 7764–7764. Used at proof lines 8804-8806 to conclude that

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the direct image of the tensor product is pseudo-coherent, which is the second condition; the deduction is inside the counted span.

Source reference. Original equation 0CS8: sources/stacks-perfect/pinned-original-source.tex, source lines 4920–4920. Cited at proof lines 8809-8811 to fix the object P used in the translation to modules; the translation argument is inside the counted span.

Source reference. Original equation 0CS9: sources/stacks-perfect/pinned-original-source.tex, source lines 4927–4927. Cited with the previous label at proof lines 8809-8811 for the algebra R; the use made of it is inside the counted span.

Source reference. Original lemma 0BQU: sources/stacks-perfect/pinned-original-source.tex, source lines 4961–4961. Invoked at proof lines 8812-8816 to obtain the equivalence and the object M mapping to K; the rest of the argument is inside the counted span.

Source reference. Original lemma 0CS6: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 8818-8822 for the displayed isomorphism; the use of it to reach the fifth condition is inside the counted span.

Source reference. Original lemma 0CSF: sources/stacks-perfect/pinned-original-source.tex, source lines 8687–8687. Invoked at proof lines 8855-8860 to conclude that the module M is pseudo-coherent, hence so is K; the identification of the derived Hom that makes it applicable is inside the counted span.

Source reference. Original section 09KK: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer in the algebra side of the argument; the descent itself is inside the counted span.

Source reference. Original lemma 09LX: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer used alongside the previous one; the use made of it is inside the counted span.

Lemma 0.2. *Let A be a ring. Let $n \geq 0$. Let $K \in D_{QCoh}(\mathcal{O}_{\mathbf{P}^n_A})$. The following are equivalent*

1. *K is pseudo-coherent,*
2. *$R\Gamma(\mathbf{P}^n_A, E \otimes^{\mathbf{L}} K)$ is a pseudo-coherent object of $D(A)$ for each pseudo-coherent object E of $D(\mathcal{O}_{\mathbf{P}^n_A})$,*
3. *$R\Gamma(\mathbf{P}^n_A, E \otimes^{\mathbf{L}} K)$ is a pseudo-coherent object of $D(A)$ for each perfect object E of $D(\mathcal{O}_{\mathbf{P}^n_A})$,*
4. *$R\operatorname{Hom}_{\mathbf{P}^n_A}(E, K)$ is a pseudo-coherent object of $D(A)$ for each perfect object E of $D(\mathcal{O}_{\mathbf{P}^n_A})$,*
5. *$R\Gamma(\mathbf{P}^n_A, K \otimes^{\mathbf{L}} \mathcal{O}_{\mathbf{P}^n_A}(d))$ is pseudo-coherent object of $D(A)$ for $d = 0, 1, \ldots, n$.*

Proof. Recall that

$$R\operatorname{Hom}_{\mathbf{P}^n_A}(E, K) = R\Gamma(\mathbf{P}^n_A, R\mathcal{H}om_{\mathcal{O}_{\mathbf{P}^n_A}}(E, K))$$

by definition, see Cohomology, Section 0B6A. Thus parts (4) and (3) are equivalent by Cohomology, Lemma 08DQ.

Since every perfect complex is pseudo-coherent, it is clear that (2) implies (3).

Assume (1) holds. Then $E \otimes^{\mathbf{L}} K$ is pseudo-coherent for every pseudo-coherent E, see Cohomology, Lemma 09J3. By Lemma 0CSD the direct image of such a pseudo-coherent complex is pseudo-coherent and we see that (2) is true.

Part (3) implies (5) because we can take $E = \mathcal{O}_{\mathbf{P}^n_A}(d)$ for $d = 0, 1, \ldots, n$.

To finish the proof we have to show that (5) implies (1). Let P be as in (0CS8) and R as in (0CS9). By Lemma 0BQU we have an equivalence

$$- \otimes^{\mathbf{L}}_R P : D(R) \longrightarrow D_{QCoh}(\mathcal{O}_{\mathbf{P}^n_A})$$

Let $M \in D(R)$ be an object such that $M \otimes^{\mathbf{L}} P = K$. By Differential Graded Algebra, Lemma 0CS6 there is an isomorphism

$$R\operatorname{Hom}(R, M) = R\operatorname{Hom}_{\mathbf{P}^n_A}(P, K)$$

in $D(A)$. Arguing as above we obtain

$$R\operatorname{Hom}_{\mathbf{P}^n_A}(P, K) = R\Gamma(\mathbf{P}^n_A, R\mathcal{H}om_{\mathcal{O}_{\mathbf{P}^n_A}}(E, K)) = R\Gamma(\mathbf{P}^n_A, P^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_{\mathbf{P}^n_A}} K).$$

Using that $P^\vee$ is the direct sum of $\mathcal{O}_{\mathbf{P}^n_A}(d)$ for $d = 0, 1, \ldots, n$ and (5) we conclude $R\operatorname{Hom}(R, M)$ is pseudo-coherent as a complex of A-modules. Of course $M = R\operatorname{Hom}(R, M)$ in $D(A)$. Thus M is pseudo-coherent as a complex of A-modules. By Lemma 0CSF we may represent M by a bounded above complex $F^\bullet$ of finite free R-modules. Then $F^\bullet = \bigcup_{p \geq 0} \sigma_{\geq p} F^\bullet$ is a filtration which shows that $F^\bullet$ is a differential graded R-module with property (P), see Differential Graded Algebra, Section 09KK. Hence $K = M \otimes^{\mathbf{L}}_R P$ is represented by $F^\bullet \otimes_R P$ (follows from the construction of the derived tensor functor, see for example the proof of Differential Graded Algebra, Lemma 09LX). Since $F^\bullet \otimes_R P$ is a bounded above complex whose terms are direct sums of copies of P we conclude that the lemma is true. □